Exercise Sheet 4: Vector Spaces

October-November 2024

1. Let $u = (1, 2, 3)$ and $v = (-1, 3, 4)$, find $3u - 2v$.

2. Solve for x provided that $u = (1, -1, 2)$, $v = (0, 2, 3)$ and $w = (0, 1, 1)$ and

$$2x - u + 3v + 2w = 0.$$

3. Determine which sets together with the two operations (scalar multiplication and addition) defined respectively are vector spaces:

   (a) The set of all triples of real numbers (x, y, z) with the following operations $(x, y, z)+(x', y', z') = (x + x', y + y', z + z')$ and $c \cdot (x, y, z) = (cx, y, z)$;

   (b) The set of all pairs of real numbers (x, y) with the following operations $(x, y) + (x', y') = (x + x', y + y')$ and $c \cdot (x, y) = (2cx, 2cy)$;

   (c) The set of all pairs of real numbers of the form (x, y) where $x \geq 0$, with the standard operations on $\mathbb{R}^2$;

   (d) All 2 by 2 matrices with the matrix addition and scalar multiplication;

   (e) The set of singular 2 by 2 matrices with the matrix addition and scalar multiplication;

   (f) The set of non-singular 2 by 2 matrices with the matrix addition and scalar multiplication;

   (g) The set of all 2 by 2 matrices of the form $\begin{bmatrix} a & 1 \\ 1 & b \end{bmatrix}$ with the matrix addition and scalar multiplications;

   (h) All one variable polynomials of degree 2 with the scalar multiplication and addition are the scalar multiplication and additions in polynomials.

4. Which of the following subsets of $\mathbb{R}^3$ are actually subspaces?

   (a) The plane of vectors (b_1, b_2, b_3) with first component $b_1 = 0$.

   (b) The plane of vectors b with $b_1 = 1$.

   (c) The vectors b with $b_2 b_3 = 0$ (this is the union of two subspaces, the plane $b_2 = 0$ and the plane $b_3 = 0$).

   (d) All combinations of two given vectors $(1, 1, 0)$ and $(2, 0, 1)$.

   (e) The plane of vectors (b_1, b_2, b_3) that satisfy $b_3 - b_2 + 3b_1 = 0$.

5. Write v as a linear combination of u_1, u_2 and u_3 if possible, where

$$v = (3, 0, -6),\ u_1 = (1, -1, 2),\ u_2 = (2, 4, -2),\ u_3 = (1, 2, -4).$$

6. Determine whether S (a) spans $\mathbb{R}^3$, (b) is linearly independent, and (c) is a basis for $\mathbb{R}^3$, where
$$S = \{(2, 0, -1), (2, -1, 1), (4, 2, 0)\}.$$

7. Determine whether $S = \{1 - x, 2x + 3x^2, x^2 - 2x^3, 2 + x^3\}$ is a basis for $P_3(x)$.

8. Determine whether the following set S is a basis for $M_{2\times 2}$, where
$$S = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} -1 & 0 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \right\}.$$

9. Find a basis for the solution space of $Ax = 0$, where

   a) $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$,

   b) $A = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$,

   c) $A = \begin{bmatrix} 1 & 0 & -2 & 0 \\ 4 & -2 & 4 & -2 \\ -2 & 0 & 1 & 3 \end{bmatrix}$.

10. Find the rank, a basis for the row space and a basis for the column space of the matrix A:

    a) $A = \begin{bmatrix} 1 & 2 & -1 \end{bmatrix}$,

    b) $A = \begin{bmatrix} 1 & 2 \\ -4 & 3 \\ 6 & 1 \end{bmatrix}$,

    c) $A = \begin{bmatrix} 1 & 2 & 0 \\ -1 & 4 & 1 \\ 0 & 1 & 3 \end{bmatrix}$.

11. Find a basis for the subspace of $\mathbb{R}^3$ spanned by S, where
$$S = \{(1, 2, 2), (-1, 0, 0), (1, 1, 1)\}.$$

12. Find a basis and the dimension of the solution space of the homogeneous system of linear equations
$$\begin{aligned} x_1 + 3x_2 + 2x_3 + 22x_4 + 13x_5 &= 0 \\ x_1 + 2x_3 - 2x_4 + x_5 &= 0 \\ 3x_1 + 6x_2 + 5x_3 + 42x_4 + 27x_5 &= 0. \end{aligned}$$

13. Determine whether the nonhomogeneous system $Ax = b$ is consistent, and if the system is consistent, write the solution in the form $x = x_h + x_p$, where x_h is a solution of $Ax = 0$ and x_p is a particular solution of $Ax = b$:
$$\begin{aligned} x_1 + 2x_2 + x_3 + x_4 + 5x_5 &= 0 \\ -5x_1 - 10x_2 + 3x_3 + 3x_4 + 55x_5 &= -8 \\ x_1 + 2x_2 + 2x_3 - 3x_4 - 5x_5 &= 14 \\ -x_1 - 2x_2 + x_3 + x_4 + 15x_5 &= -2. \end{aligned}$$

14. Let $B = \{(1,1),(-1,1)\}$ be a basis in $\mathbb{R}^2$ and let x be a vector in $\mathbb{R}^2$ with $[x]_B = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$. Find the coordinate matrix of x relative the standard basis.

15. Find the coordinate matrix of x relative to the (non-standard) basis B of $\mathbb{R}^m$:

    (a) $B = \{(1,1),(0,-2)\}$, $x = (2,-1)$.

    (b) $B = \{(1,0,0),(0,1,0),(1,1,1)\}$, $x = (4,-2,9)$.

    (c) $B = \{(1,2,3),(1,2,0),(0,-6,2)\}$, $x = (3,-3,0)$.

16. Find the coordinate matrix of x relative to the basis B', where

    (a) $B = \{(1,1),(1,-1)\}$, $B' = \{(0,1),(1,2)\}$; $[x]_B = \begin{bmatrix} 3 \\ -3 \end{bmatrix}$,

    (b) $B = \{(1,0,0),(1,1,0),(1,1,1)\}$, $B' = \{(0,0,1),(0,1,1),(1,1,1)\}$; $[x]_B = \begin{bmatrix} -1 \\ 2 \\ -3 \end{bmatrix}$.

17. Find the transition matrix from B to B':

    (a) $B = \{(1,-1),(3,1)\}$, $B' = \{(1,2),(-1,0)\}$.

    (b) $B = \{(1,1,1),(1,1,0),(1,0,0)\}$, $B' = \{(1,2,3),(0,1,0),(1,0,1)\}$.

18. Let v_1, v_2 and v_3 be three linearly independent vector in a vector space V. Is the set $\{v_1 - v_2, v_2 - v_3, v_3 - v_1\}$ linearly dependent or linearly independent?